Practical - 8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

A . Theory

Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

1. Is the schema normalized properly?

Yes, the schema is normalized properly. I have divided the data into different tables like students, events and registrations. This reduces duplicate data and makes the database easier to manage.

2. Are primary keys and foreign keys correctly defined?

Yes, primary keys and foreign keys are correctly defined. Primary keys are used to uniquely identify each record, and foreign keys are used to create a relationship between tables, like students and registrations.

3. Is the database connection handled securely?

Yes, the database connection is handled using PHP and MySQL. I have kept the database connection details in the PHP code and for user input, prepared statements can be used to prevent SQL injection.

B. Implementation/ Output/Result

Here is my GitHub repo link the codes are here in repo:
https://github.com/dhyaani19/StudentHub-Portal.git

C.. Evidence of Work (Screenshots)

Figure 8.1 : Profilee page (profile.php)

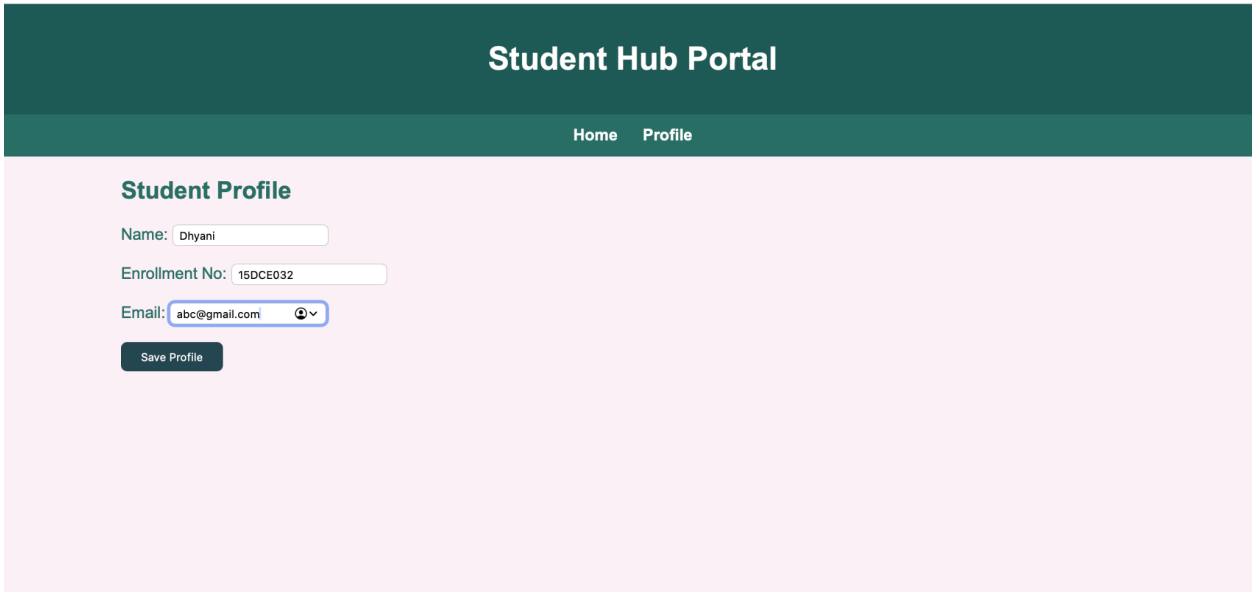


Figure 8.2 : Database events

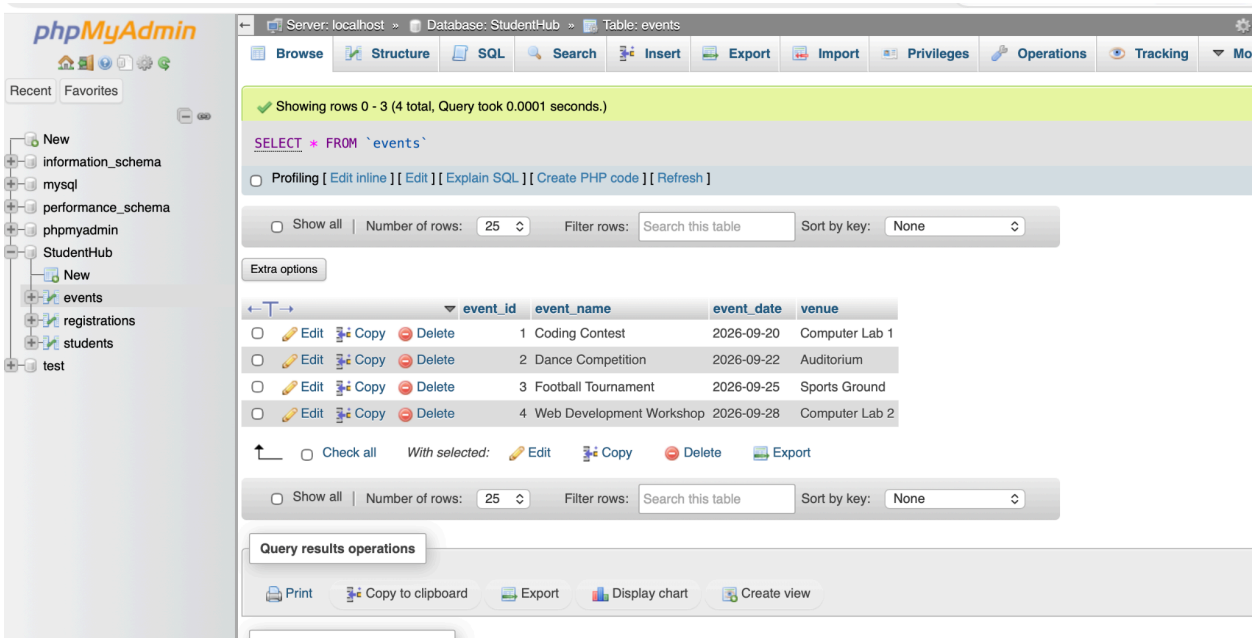
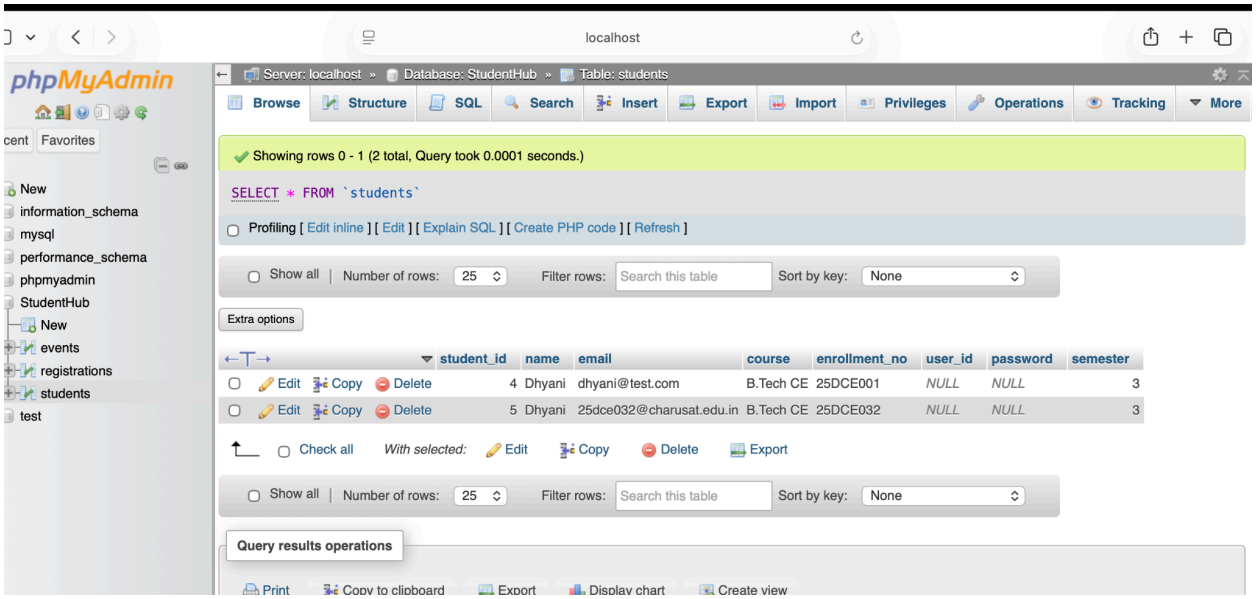


Figure 8.3 : Database student table



D. GitHub Repository Link:

https://github.com/dhyaani19/StudentHub-Portal.git

E. Challenges Faced:

F. Learning Outcomes:

In this practical, I learned how to connect an HTML form with PHP and receive form data using the POST method. I learned how to validate user inputs and store the submitted data in a file. I also understood basic file handling and how to display success and error messages to the user. Overall, this practical helped me understand how a website can take user input and process it on the server side.

Faculty Remarks: